Amrita School of Artificial Intelligence, Faridabad
Amrita Vishwa Vidyapeetham

Course Code: **23MAT214**

Mathematics for Intelligent System-4

Part- 1.7

Hankel Matrices

BTech 4th Sem

Dr. Anirban Tarafdar
Assistant Professor, School of AI
Amrita Vishwa Vidyapeetham, Faridabad

*All Sources: Stanford University, Texas Tech University, UC Santa Barbara, etc.
*Sharing or distributing this lecture material without the permission of the lecture coordinator is strictly prohibited.

Motivation

In previous lectures we studied two structured matrices:

Toeplitz matrices

•Used to represent **linear convolution**

•Appear in **linear shift-invariant systems**

Circulant matrices

•Used to represent **circular convolution**

•Diagonalized by the **Fourier transform**

Both Toeplitz and circulant matrices arise when a system processes signals.

However, in many real-world problems we want to analyze **the underlying dynamics of a time series itself**, rather than simply applying a filter.

Examples include:

- predicting stock market movements
- analyzing heart rhythms
- identifying vibration patterns in machines
- studying fluid flow patterns
- discovering hidden periodic behavior in signals

In these problems, we convert a signal into a special structured matrix called a **Hankel matrix**.

Hankel matrices are fundamental in:

•system identification

•signal reconstruction

•time-delay embedding

•dynamic mode decomposition (DMD)

•subspace methods in control theory

Definition of a Hankel Matrix

A **Hankel matrix** is a matrix whose **anti-diagonals contain the same value**.

General form:

$$H = \begin{bmatrix} h_1 & h_2 & h_3 & \dots & h_n \\ h_2 & h_3 & h_4 & \dots & h_{n+1} \\ h_3 & h_4 & h_5 & \dots & h_{n+2} \\ \vdots & \vdots & \vdots & & \vdots \end{bmatrix}$$

Key Property

Each entry depends on the **sum of its indices**:

$$H_{i,j} = h_{i+j}$$

Thus:

- entries on the same **anti-diagonal** are identical.

AMRITA VISHWA VIDYAPEETHAM DEEMED TO BE UNIVERSITY
Dr. Anirban Tarafdar
Assistant Professor, School of AI
Amrita Vishwa Vidyapeetham, Faridabad

Constructing a Hankel Matrix From a Signal

Suppose we observe a signal

$$x = (1, 2, 3, 4, 5, 6)$$

This signal may represent:

- sensor measurements
- stock prices over time
- audio amplitude values

We now convert this signal into a Hankel matrix.

Step 1: Choose window length

Let the window length be **3**.

Step 2: Build rows using shifted segments

Row 1

$$(1, 2, 3)$$

Row 2

$$(2, 3, 4)$$

Row 3

$$(3, 4, 5)$$

Row 4

$$(4, 5, 6)$$

Final Hankel matrix

$$H = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \\ 4 & 5 & 6 \end{bmatrix}$$

Real Meaning of the Hankel Matrix

The Hankel matrix stores **time-delayed copies of the signal**.

Example signal

$$x = (x_1, x_2, x_3, x_4, x_5, x_6)$$

Hankel matrix

$$H = \begin{bmatrix} x_1 & x_2 & x_3 \\ x_2 & x_3 & x_4 \\ x_3 & x_4 & x_5 \\ x_4 & x_5 & x_6 \end{bmatrix}$$

Interpretation:

Row 1 → signal at time 1–3

Row 2 → signal at time 2–4

Row 3 → signal at time 3–5

Row 4 → signal at time 4–6

Thus each row contains a **delayed snapshot of the system**.

arafdar
r, School of AI
eetham, Faridabad
Amrita

Why This Is Useful

Many systems have **hidden dynamics** that are not visible in a single signal.

By stacking delayed versions of the signal, we can reveal patterns.

Examples:

- oscillations
- trends
- growth/decay rates
- periodic behavior

AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY
Dr. Anirban Tarafdar
Assistant Professor, School of AI
Amrita Vishwa Vidyapeetham, Faridabad

Example 1: Heartbeat Analysis (ECG)
Doctors analyze electrical signals from the heart.
A single signal may look noisy.
But Hankel matrices allow:
• detection of abnormal rhythms
• prediction of heart conditions
• identification of periodic patterns
Used in **medical signal processing**.

Example 2: Fault Detection in Machines
Sensors measure vibrations in engines or turbines.
If a bearing begins to fail, the vibration pattern changes.
Hankel matrices help detect:
• periodic fault signatures
• mechanical instability
• abnormal oscillations
Used in **predictive maintenance**.

Example 3: Climate and Weather Analysis
Temperature measurements over time:

$$x(t)$$

Hankel matrices allow scientists to analyze:
• seasonal patterns
• oscillations in climate systems
• long-term trends
Used in **climate modeling**.

Example 4: Speech Processing
Speech signals are time-varying.
Hankel matrices help extract features such as:
• phoneme patterns
• pitch structure
• speech dynamics
Used in **speech recognition systems**.

AMRITA VIDYAPEETHAM
Amrita